IGSS OPC Client and Server



INSIGHT AND OVERVIEW

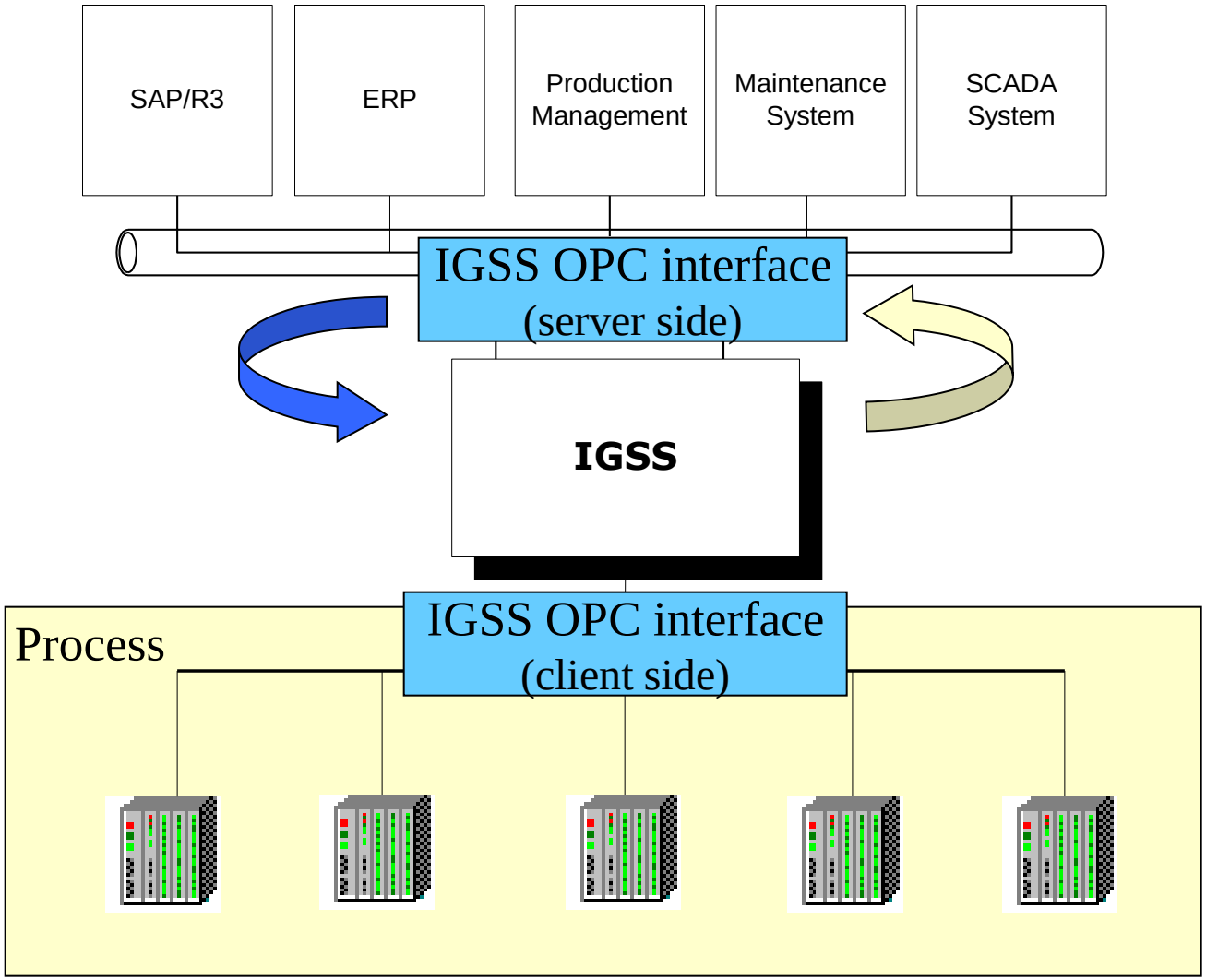
Topics

- What is OPC ?
- IGSS and OPC
- IGSS OPC Client-Side Driver
- IGSS OPC Server
- SCADA to SCADA with OPC
- Configuration af DCOM security

What is OPC ?

- OPC = OLE for Process Control
- OPC is an interface standard developed and maintained by the OPC Foundation (http://www.opcfoundation.org/)
- Winning greater acceptance (becoming industry standard)
- Integrate hardware from different manufacturers in plant (OPC Client-side Driver)
- Interface to different SCADA systems, industrial control and reporting applications (IGSS OPC Server)

IGSS and OPC

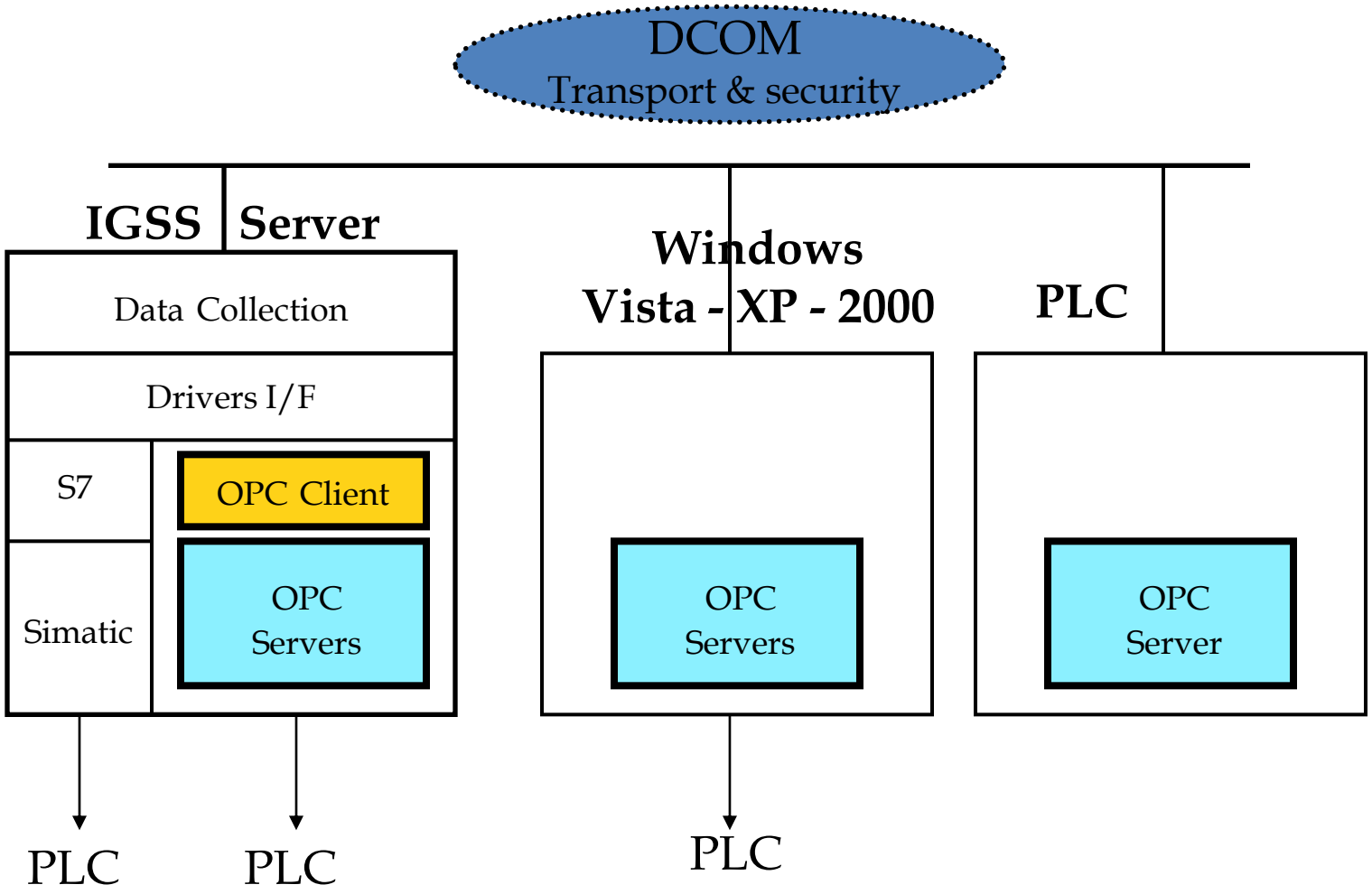

SAP/R3
ERP
Production Management
Maintenance System
SCADA System
IGSS OPC interface (server side)
IGSS
IGSS OPC interface (client side)
Process

OPC Client-Side Driver

Purpose

- The IGSS OPC client connects to an OPC Server
- The OPC Server acts as the communication channel between IGSS and the PLCs in the process

IGSS OPC Client



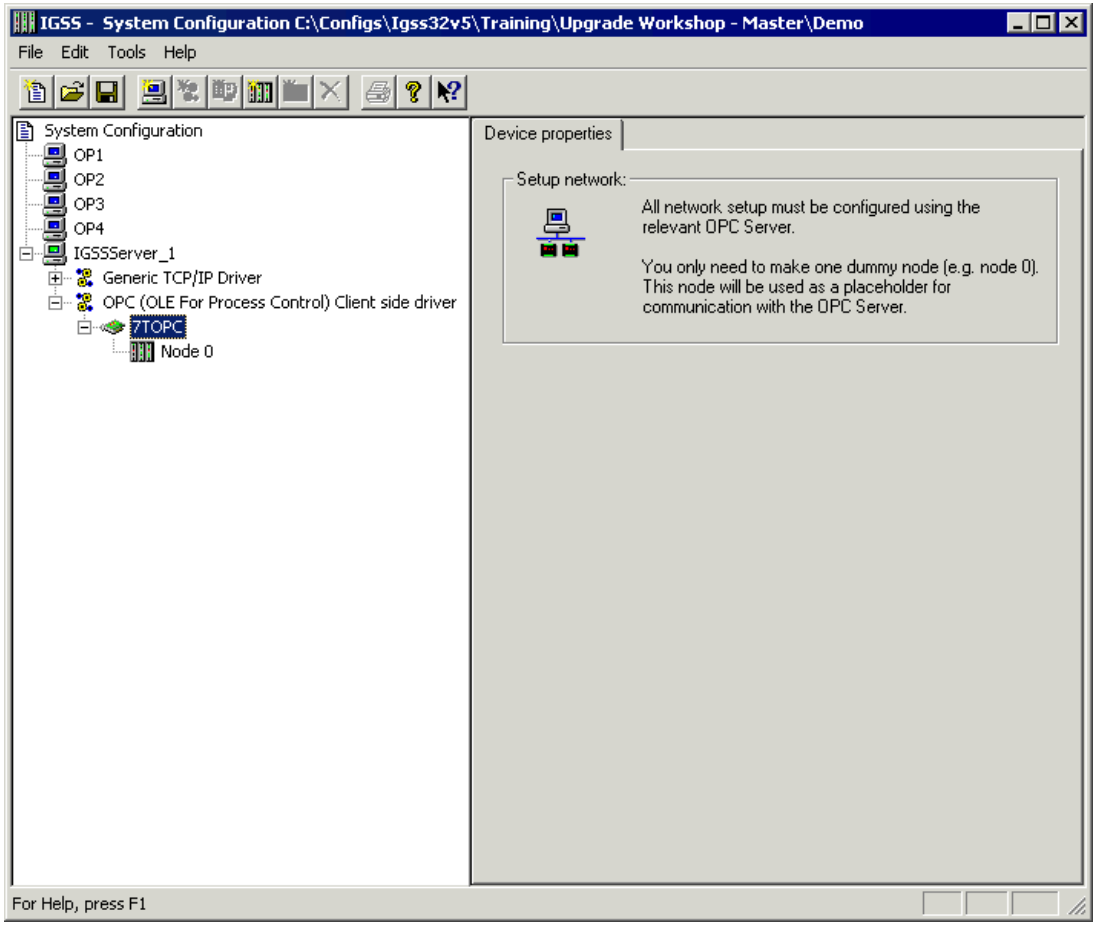
Driver History

- Released with IGSS32 Version 3, January 2001
- PLC drivers still developed by 7T but the number is decreasing due to increase in number of OPC servers commercially available

Activating the OPC driver

- Activate the OPC driver with just a single click in System Configuration.
- Definition of communication channels, nodes, addressing, etc. now found in the OPC Server itself.
- The PLC addressing of an IGSS object is different from what is characteristic for other drivers on the Edit Mapping tab.

Driver in System Configuration



Simple setup
Find driver ID = 42, OPC (OLE for Process Control) Client Side Driver

Click once and the required settings are entered in System Configuration.

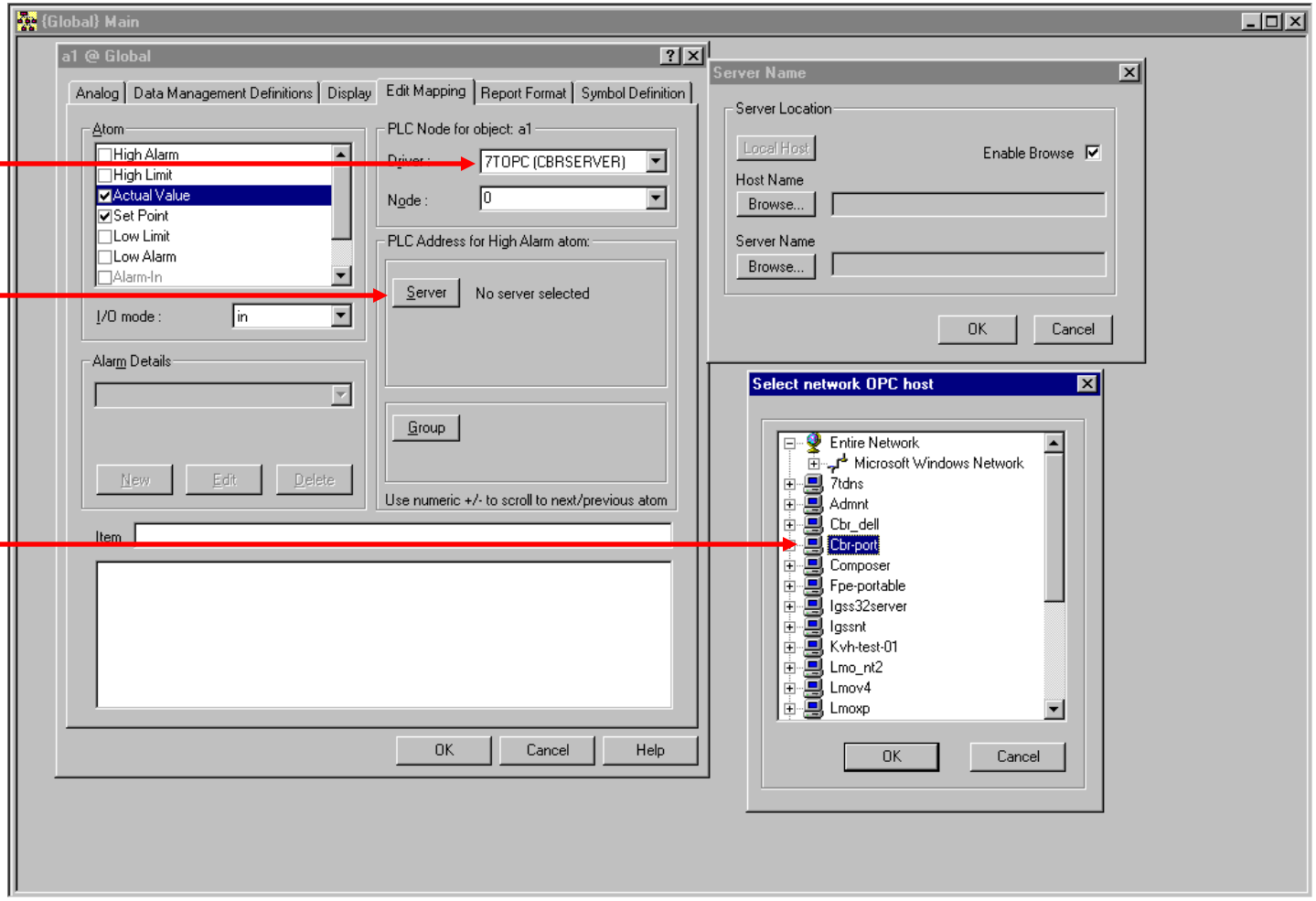
All further setup is done in the OPC Server.

Choosing the OPC Host

Choose OPC driver

Start OPC choice of server

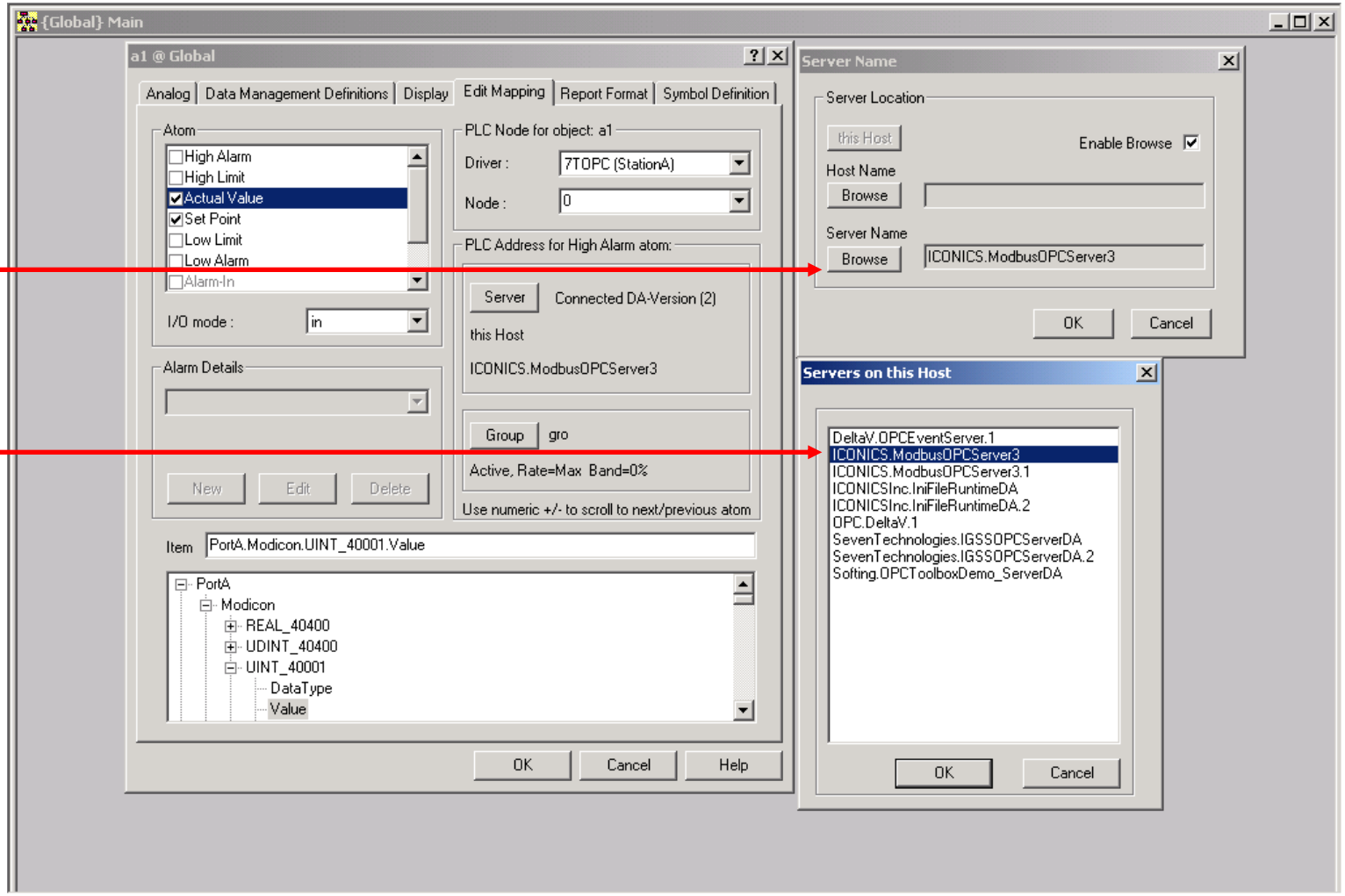
Choose host for OPC server or use Local Host



Choosing the OPC Server

Browse OPC servers

Choose OPC server



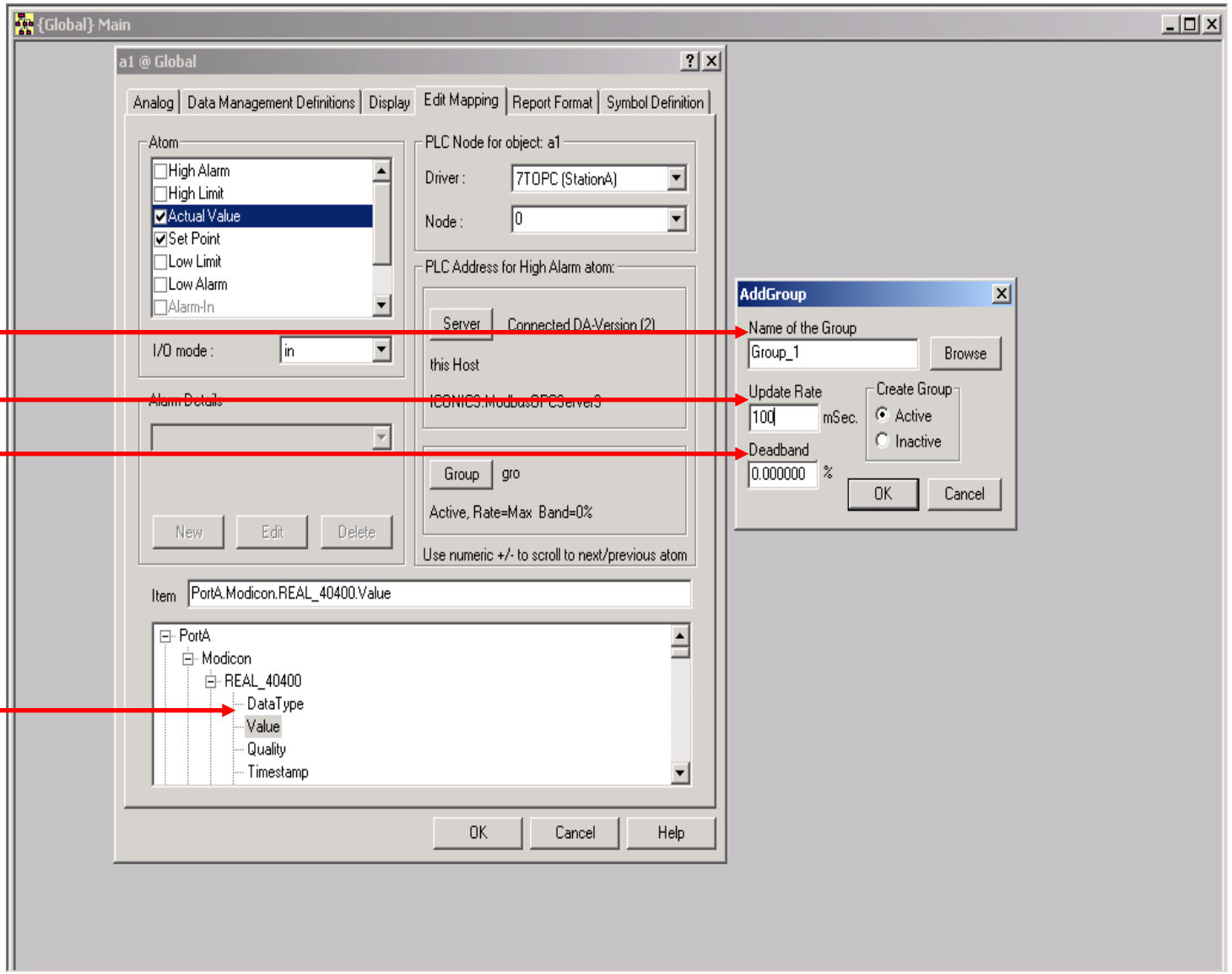
Choosing the OPC Group & Item

Group Name

Update Rate

Deadband

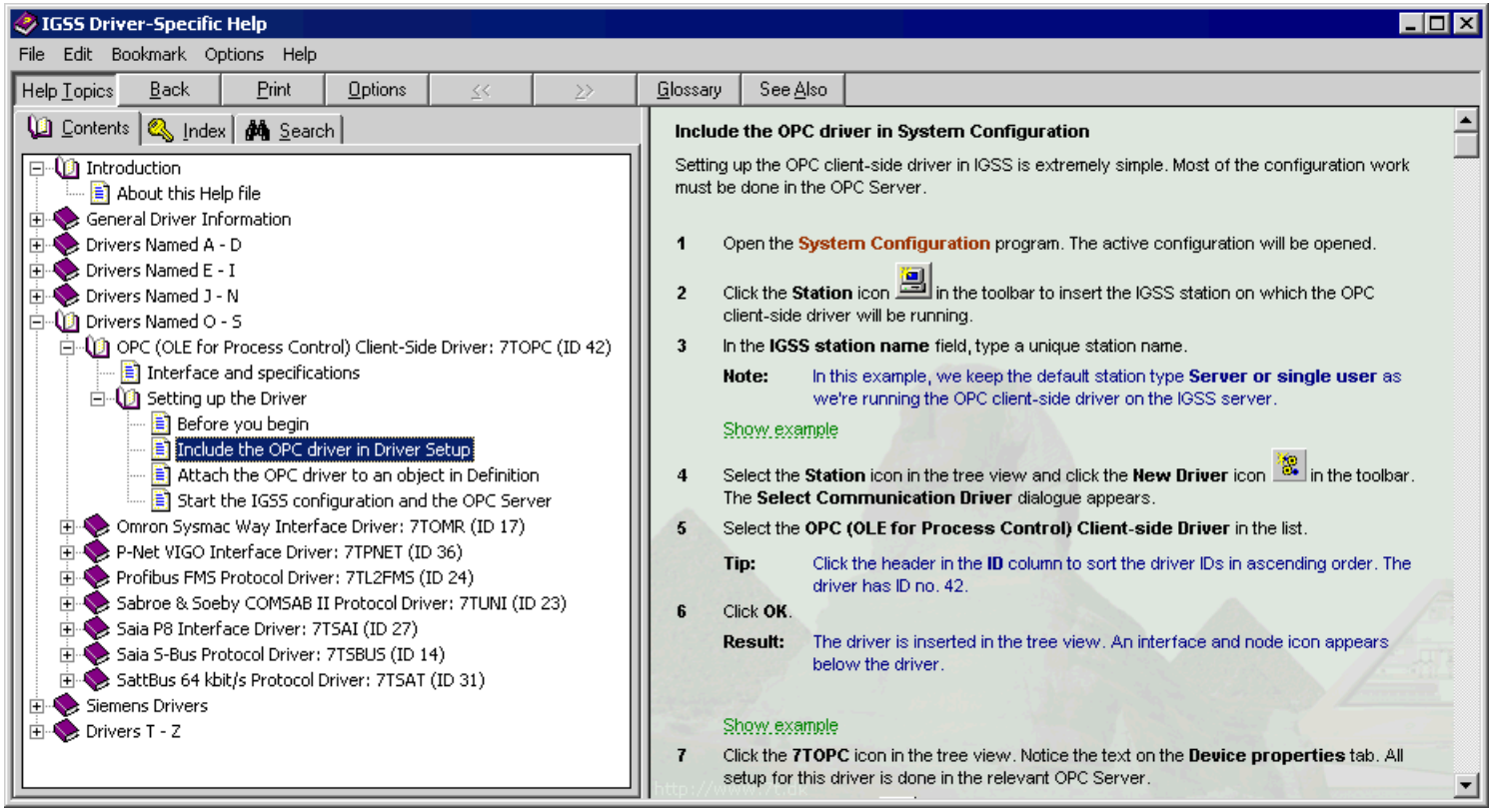
Choose OPC Item



Help on OPC Client

Driver-Specific Help

Activate the help file from System Configuration or
from the ”IGSS Help Files” subfolder of the IGSS program group.



IGSS OPC Server

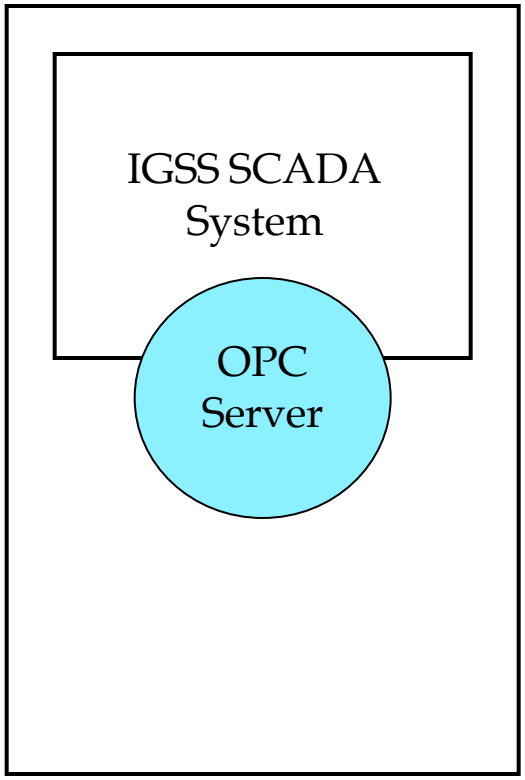
Purpose

- Allows any OPC client to connect to IGSS with the purpose of reading or writing values to the IGSS configuration
- Typical scenarios:
  - SCADA to SCADA data exchange
  - Use IGSS values for administrative calculations

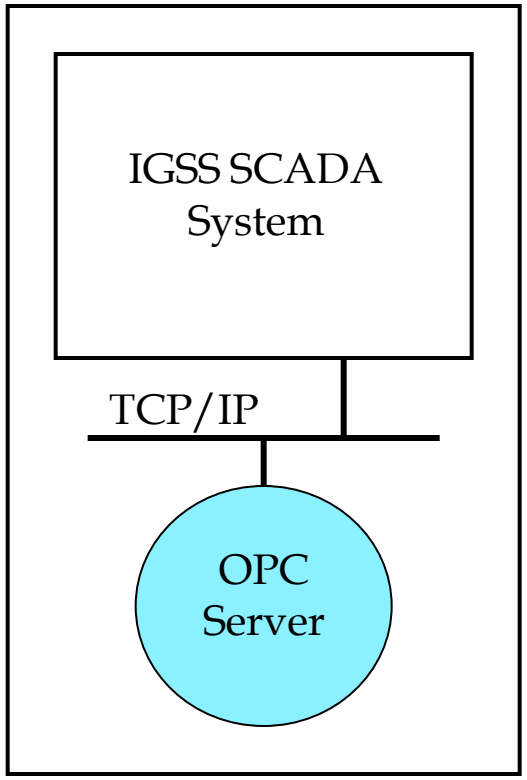
IGSS OPC Server History

- The OPC server is not appended to existing code but connected with a TCP/IP interface.
- Configuration saved in ACCESS too slow.
- Beta test at Arla Foods.
- Release 1 with IGSS32 Version 4 March 2002.

Integration of OPC Server

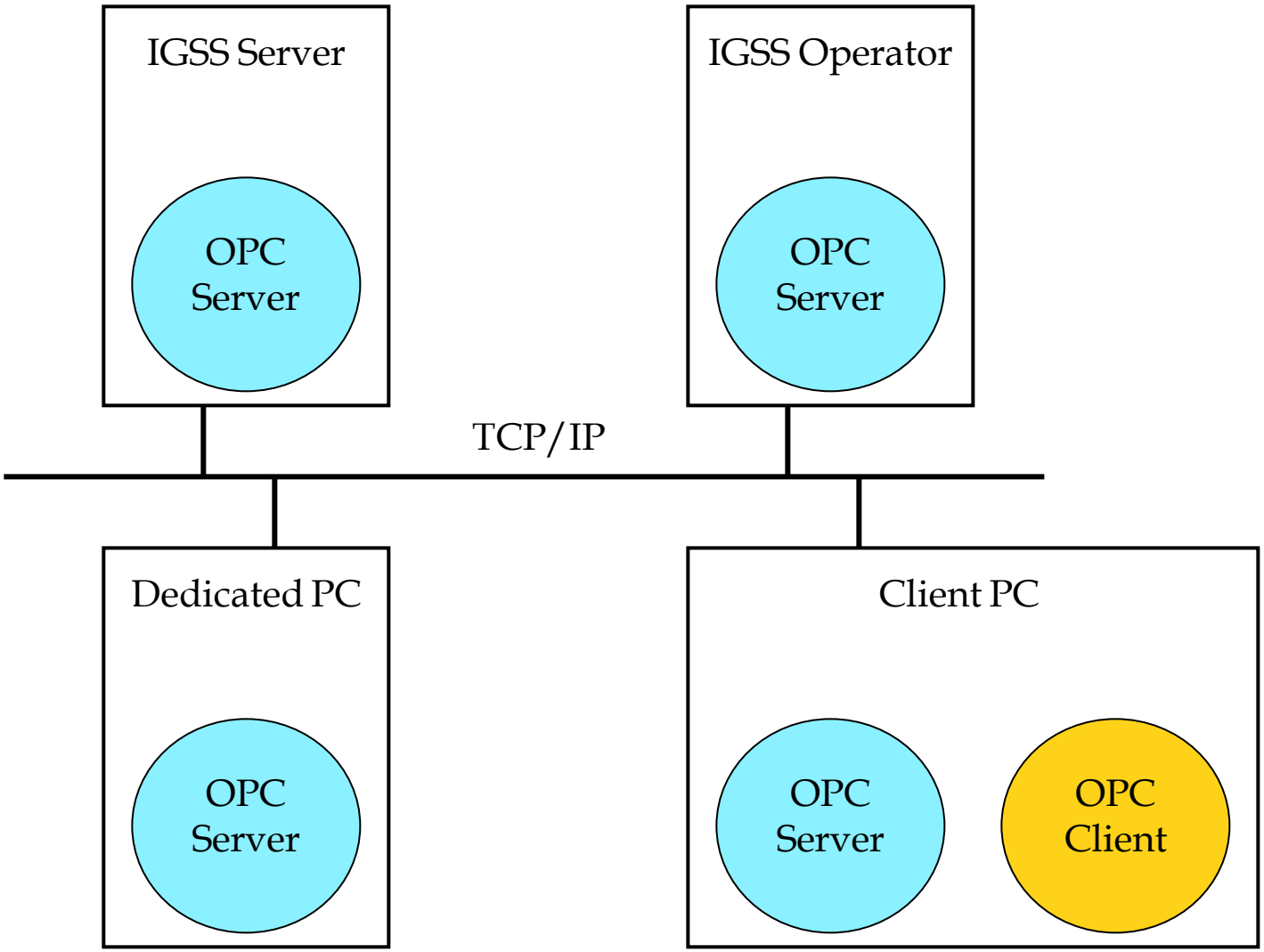


Limited flexibility



The solution chosen

OPC Server Deployment

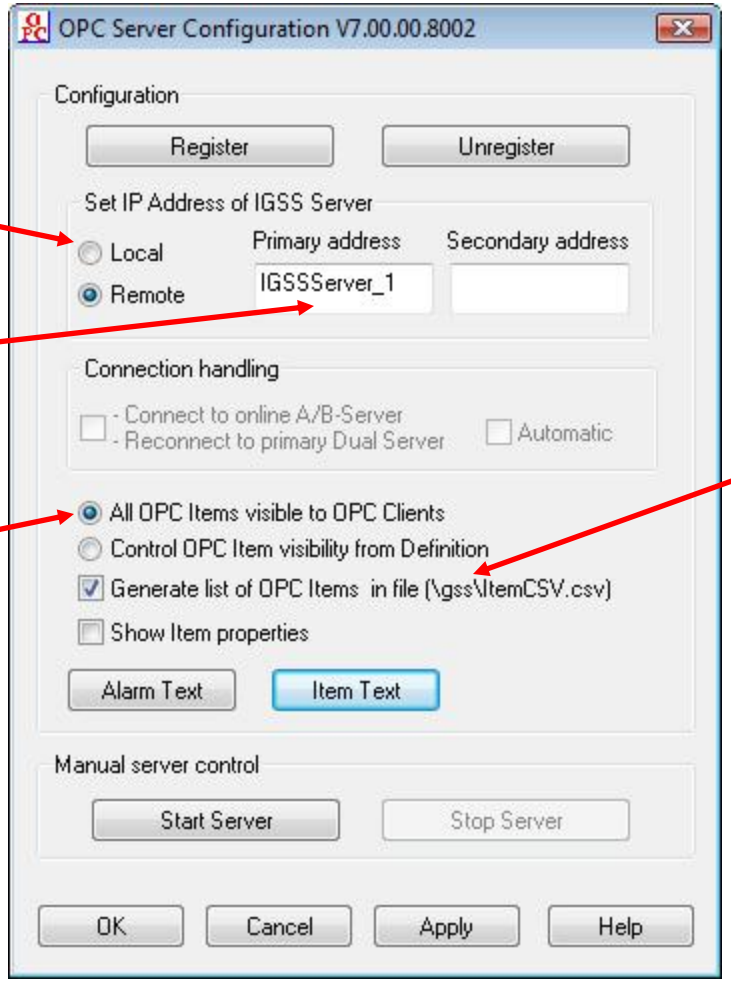


Setup of OPC Server

OPC server and IGSS server on same machine

IP address of primary (and secondary) IGSS server

Should all IGSS objects be visible ?

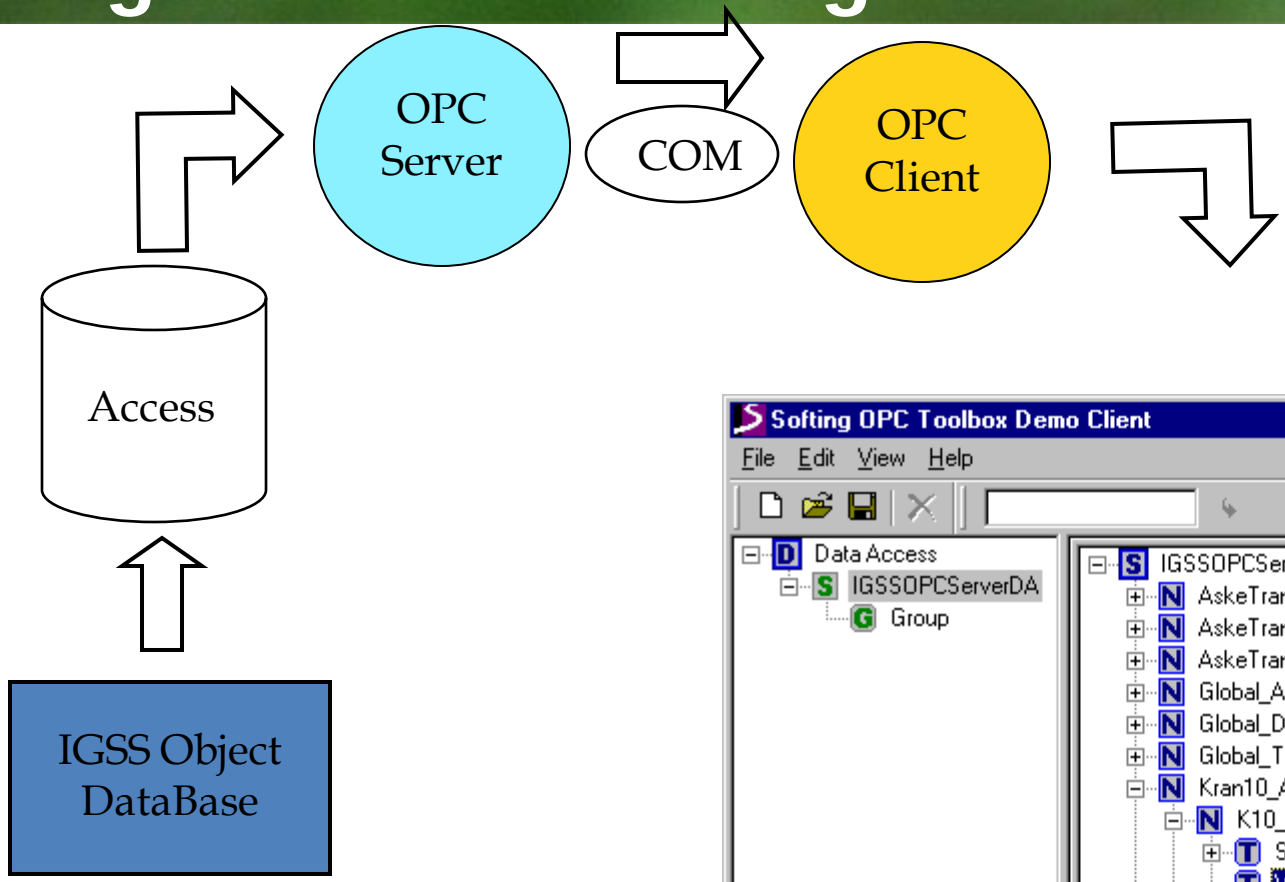


If enabled, a .csv file is generated listing all objects in the configuration.

The list will be generated when the first client connects.

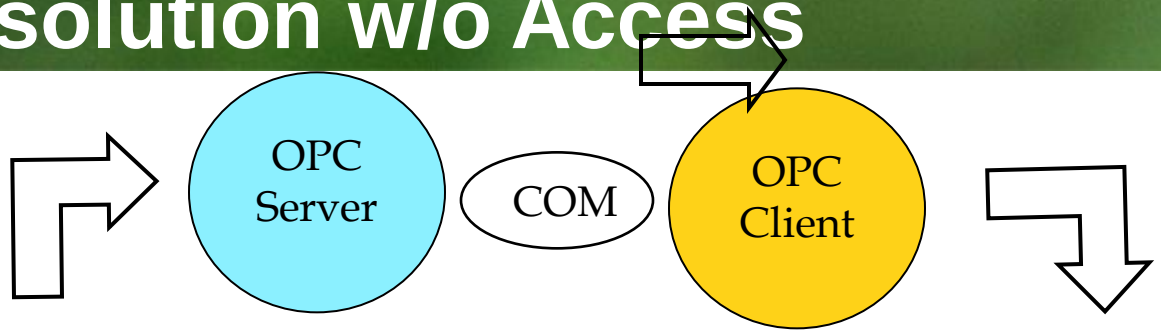
Can be imported into the OPC Client as OPC items.

Using Access as config. tool



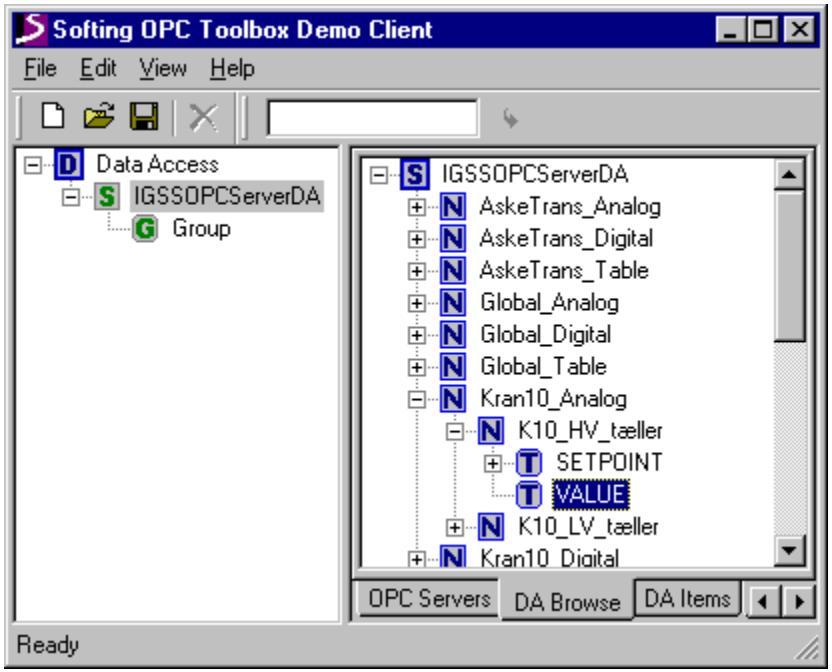
With Access there's a 12 minute lapse before 10,000 elements are visible in the OPC client browse window

Final solution w/o Access



IGSS Object DataBase

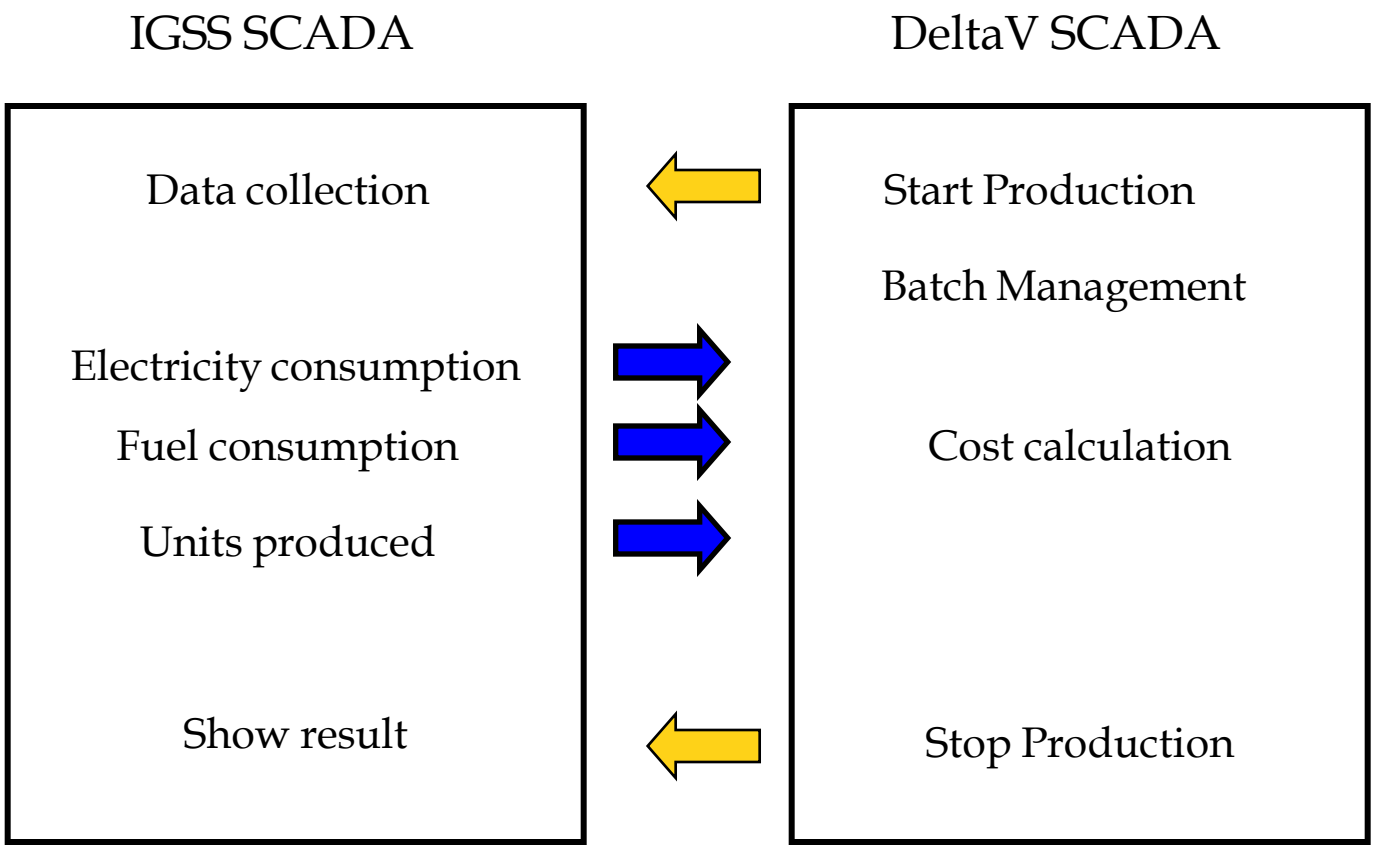
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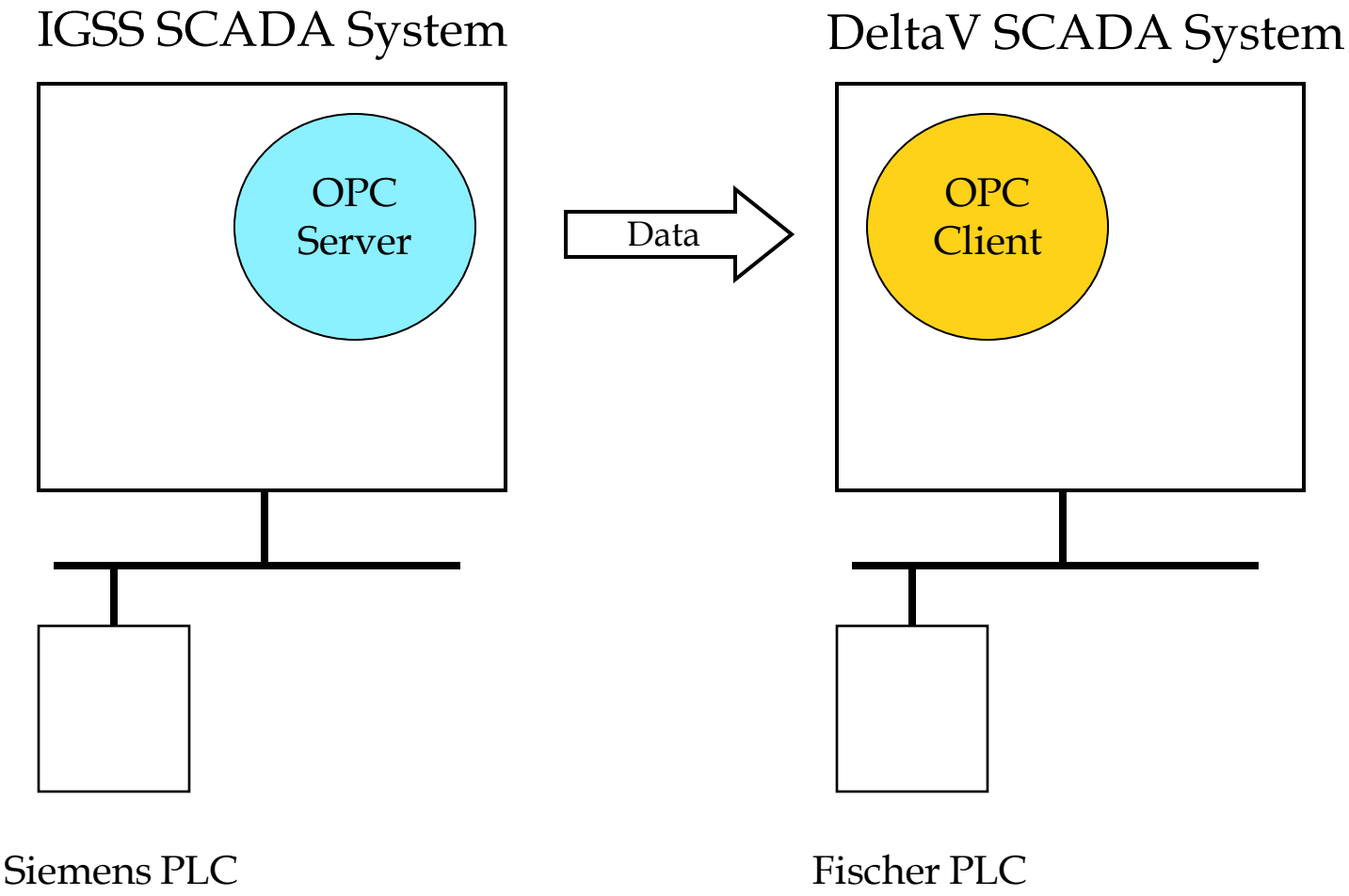
Arla Foods

- Needed data exchange between two SCADA systems: DeltaV and IGSS
- Primary data flow from IGSS to DeltaV.
- Preferred solution: IGSS as OPC Server and DeltaV as OPC Client.
- Valuable experience gained in configuring DCOM security.

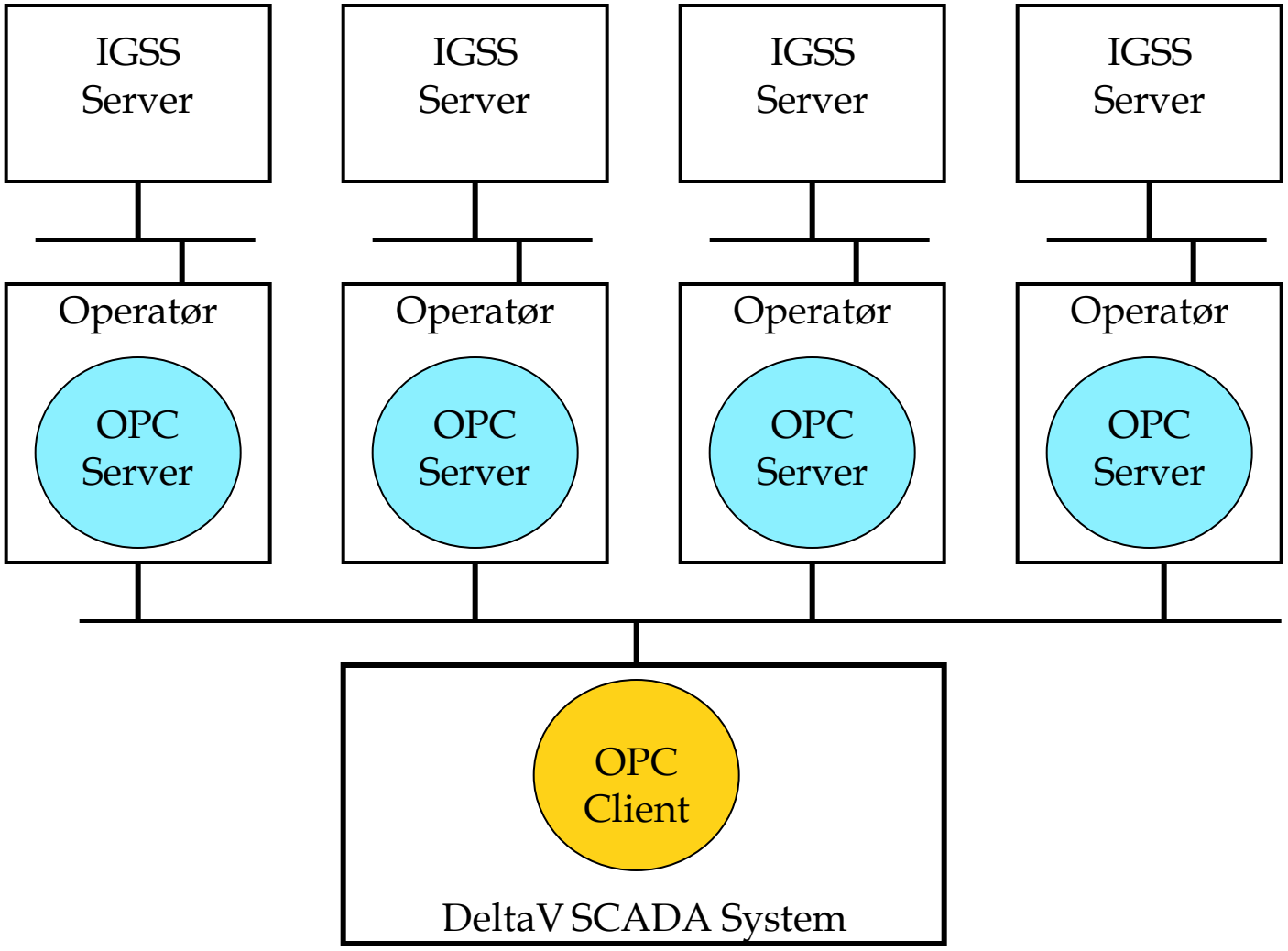
Task allocation



Final solution



Current solution at Arla Foods

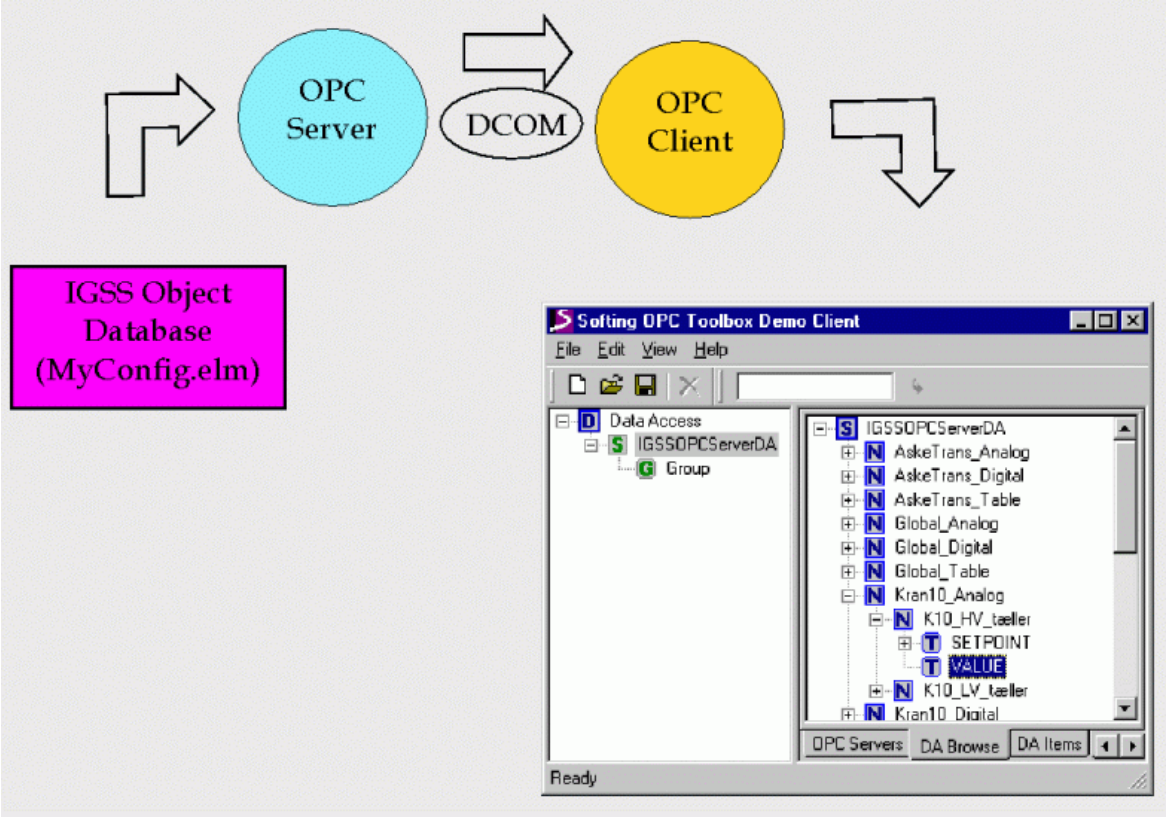


DCOM Security

- DCOM = Distributed Component Object Model
- DCOM (Distributed COM) is an integral security mechanism in Windows
- DCOM is a network extension of the COM technology
- Supports communication between objects on a network
- Enables the use of distributed programs

IGSS OPC Server uses DCOM

- When the client connects to the IGSS OPC Server, DCOM is used as the security layer.



The DCOM Configuration Tool

- DCOM security settings are defined in DCOM Configuration Properties
- Choose Start > Run > Dcomcnfg
- Two sets of security settings:
  - Default settings for all DCOM applications
  - Specific settings for the individual DCOM application
- The IGSS ”Interface Help” file describes the latter method

Help on IGSS OPC Server

Interface Help

Opened from the IGSS Start menu.

Go to the book ”IGSS OPC Server”

